Weight-aware imputation for policy microsimulation

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Abstract

Policy microsimulation models rely on imputation to attach variables one survey observes to records from another, and the imputed file is then reweighted downstream: calibrated to administrative totals, filtered to subpopulations, stressed under reform scenarios. Fitting imputation models to survey records while ignoring their design weights biases imputed conditionals toward the sample rather than the population. We present the imputation operator of MICROCOSM, POLICYENGINE's open-source microdata stack: a regime-gated, sequentially chained, weighted-bootstrap quantile-regression-forest estimator whose interface makes an unweighted fit impossible to request by accident. To evaluate it we formalize a *population-view harness*: one latent population observed through survey views, with any candidate weighted file scored by projecting it through each view against that survey's holdout. Benchmarks against standard methods on four task families knock out each design choice. Fit without weights, the identical estimator is six times worse on net-worth marginal fit and imputes a 99th percentile 1.9 to 2.3 times the holdout's — an inflation that capped joint-geometry metrics miss and an uncapped tail axis detects. With its gates off, it misses the dividend zero share by 7.2 percentage points, not 0.6. Fitting four balance-sheet components independently costs 5.8 points of classifier two-sample AUC; the accounting identity recovers them, and chaining adds 1.2. A hot deck that draws donors in proportion to weight matches the estimator on marginal fit within a survey and trails it by 1.6 to 2.9 AUC points across surveys. The estimator is available standalone as `microcosm-fit`, and every number regenerates from the repository.

Keywords: microsimulation, imputation, survey weights, quantile regression forests, statistical matching

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1 Introduction

Microsimulation models of tax and transfer policy require individual records that span domains no single survey covers (Bourguignon and Spadaro, 2006; Sutherland and Figari, 2013). The Current Population Survey observes employment and income but not wealth; the Survey of Consumer Finances observes wealth in detail but lacks the CPS's coverage and program detail; administrative tax records observe income components precisely but only for filers, and with narrow demographics. Model builders bridge these gaps with imputation: fit a conditional model of the missing variables on a donor survey, then draw values for the receiver's records (Little and Rubin, 2002; D'Orazio et al., 2006).

Two properties of this setting distinguish it from generic missing-data problems. First, both files are *weighted samples*: each record stands in for a different number of population units, and the quantity of interest is a population distribution, not a sample one. An imputation model fit to unweighted records recovers the sample conditional, which differs from the population conditional exactly when the design (or a prior calibration) correlates weights with outcomes. Second, the imputed file is not an endpoint. It is *reweighted downstream*: calibrated to administrative margins (Deville and Särndal, 1992; Hainmueller, 2012), filtered to subpopulations, stressed under reforms. An imputation that looks acceptable under the weights it shipped with can therefore fail under weights it never saw. A rare donor record carrying an extreme value at near-zero weight is invisible in the shipped aggregates; a later reweighting that assigns it mass turns it into a distortion of the population total. We refer to such records as landmines, and we measure exposure to them directly with a reweighting stress test (Section 5.2, Table 4).

This paper presents and evaluates the imputation operator of MICROCOSM,[1] POLICYENGINE's open-source microdata stack (PolicyEngine, 2026). The estimator combines three design choices, each aimed at one of the failure modes above. A *weighted bootstrap* materializes survey weights into the training data of each forest, so the learned conditionals are population conditionals. *Regime gates* classify each record's sign class (negative, zero, positive) before magnitudes are modeled, so zero-inflated and sign-mixed variables — ubiquitous in economic microdata (Mullahy, 1986; Lambert, 1992) — keep their mass structure. *Sequential chaining* conditions each imputed variable on the ones already drawn, so joint structure among targets survives the imputation (Raghunathan et al., 2001;

[1] MICROCOSM was first released under the name Populace. Artifact and release identifiers created before the rename keep the legacy `populace` prefix, and the reproducibility pins in the data-availability section reference the repository at a pre-rename commit.

Van Buuren and Groothuis-Oudshoorn, 2011). The interface enforces the first choice: fits are weighted by construction, and an unweighted fit must be requested explicitly.

We make three contributions:

1. **A weight-aware imputation estimator, available standalone.** We describe the estimator and its interface contract, and release it as `microcosm-fit`, installable and usable on plain data frames independently of the rest of the MICROCOSM stack.

2. **The population-view harness.** We frame the evaluation question as one latent population observed through survey *views* — variable subsets, designs, measurement idioms — and score any candidate weighted file by projecting it through each view against that survey's holdout. The score has four blocks: a strictly proper joint score (Gneiting and Raftery, 2007; Székely and Rizzo, 2013), a support-based coverage axis that is invariant to reweighting of the candidate by definition (Naeem et al., 2020), a weighted classifier two-sample test, and an uncapped tail block on imputed columns. We added the tail block after finding that capped sample-geometry metrics are blind to twofold errors in the extreme tail, where economic variables carry their policy weight. The harness works for any generator, is non-self-referential (holdouts appear nowhere upstream), and cleanly separates what record generation must get right from what calibration can later change.

3. **A benchmark against standard practice with attribution, plus a reweighting stress test.** We compare the estimator against unweighted and weighted quantile regression forests (Meinshausen, 2006), ordinary least squares imputation (von Hippel, 2013), linear quantile regression (Koenker and Bassett, 1978), and distance hot-deck statistical matching, with the donor chosen with and without its weight (Andridge and Little, 2010; D'Orazio et al., 2006), under a paired, repeated holdout protocol. Ablations knock out one design choice at a time, so weighting, gating, and chaining each get their own attribution. A closed-form fragility diagnostic measures each method's worst-case single-record exposure under bounded reweightings of the imputed file — the landmine failure mode.

The policy stakes are concrete. Supplemental Security Income conditions eligibility on countable resources (Social Security Administration, 2024), so simulating its baseline caseload on the CPS — or reforms such as the SSI Savings Penalty Elimination Act, which would raise the resource limits (119th United States Congress, 2025) — is impossible without imputed wealth. The composition of the simulated caseload is exactly a functional

of the imputed demographics–wealth joint. This paper does not simulate the SSI caseload; it measures the property any such simulation depends on. The population-view experiment of Section 4.2 scores the imputed joint directly: it asks whether households assembled from CPS demographics and imputed wealth are distributionally exchangeable with households the SCF actually observed.

The rest of the paper proceeds as follows. Section 2 situates the estimator in the survey-imputation literature. Section 3 specifies the estimator, the ablation design, the metrics, and the protocol. Section 4 describes the four benchmark tasks and their data. Section 5 reports results, Section 6 discusses limitations, and Section 7 concludes.

2 Background

2.1 Imputation methods in survey practice

Survey agencies and microsimulation teams draw on a small set of imputation families. *Hot-deck and statistical matching* methods donate observed values from a matched donor record; they preserve marginal distributions by construction because every imputed value is a real donor value (Andridge and Little, 2010; D'Orazio et al., 2006), and remain the standard data-fusion tool in European microsimulation practice (Sutherland and Figari, 2013). *Regression-based* methods impute from a fitted conditional model — ordinary least squares with residual draws (von Hippel, 2013), linear quantile regression (Koenker and Bassett, 1978; Koenker, 2005) — and extend to multiple imputation for uncertainty propagation (Rubin, 1987; Kennickell, 1998). *Tree-ensemble* methods, in particular quantile regression forests (Meinshausen, 2006; Breiman, 2001), estimate full conditional distributions non-parametrically. They are the default in POLICYENGINE's data pipelines (Ghenis, 2018; Woodruff and Ghenis, 2024) because they capture nonlinear structure without per-variable specification. The `microimpute` package (PolicyEngine, 2025) implements these families under one interface and selects among them by cross-validation; its central finding — no family dominates across datasets — motivates empirical comparison on each new task rather than a prescribed method.

2.2 Chained imputation and joint structure

Imputing several variables one at a time, each conditional only on shared predictors, destroys the dependence among imputed variables; sequential (chained-equations) imputation conditions each variable on those already imputed, retaining joint structure (Raghunathan et al.,

2001; Van Buuren and Groothuis-Oudshoorn, 2011). In data fusion the same issue appears as the conditional independence assumption of statistical matching: matched files preserve marginals but attenuate cross-source correlations (D'Orazio et al., 2006). Synthetic-data generation faces the identical trade-off at whole-file scale (Rubin, 1993).

2.3 Weights in imputation

Design weights enter imputation practice unevenly. The model-based literature often treats weights as a nuisance — if the model is correctly specified and the design variables are conditioned on, weighting is unnecessary — while the design-based tradition weights every estimator (Little and Rubin, 2002). In production microdata pipelines the model is never correctly specified and the design variables are never fully available, so the choice is consequential: an unweighted fit targets the sample conditional, and any correlation between weights and outcomes within predictor cells propagates into the imputed population distribution. Reiter et al. (2006) show by simulation that multiple imputation which ignores the design can be severely biased when the design features are related to the survey variables, and that controlling for those features in the imputation model reduces the bias.

Where the design cannot be modeled, it can be undone by resampling. Inverse sampling draws subsamples that have the structure of simple random samples (Hinkins et al., 1997). The weighted finite population Bayesian bootstrap generates synthetic populations in which the unequal selection probabilities are inverted (Dong et al., 2014), so that imputation methods built for independent, identically distributed data can be applied to them (Zhou et al., 2016). The weighted bootstrap of Section 3 is a simple relative of these methods: donor rows are redrawn with probability proportional to weight before each forest is grown. It is one resample per forest and carries none of the finite-population machinery those methods use for inference, because the product here is a drawn file, not a variance estimate.

Tree ensembles add a mechanical reason to resample: a fully-grown forest's *predictive distribution* is a leaf-membership distribution, which per-record `sample_weight` arguments do not reweight. Honoring weights therefore requires materializing them into the training data. Zero-inflated targets add a second subtlety: weighting the zero/nonzero gate by resampling can delete a rare sign class outright, so the gate must be weighted directly. Weighting by resampling is therefore not new. We are not aware of prior survey-imputation work that combines it with sign-regime gating and chaining in one forest estimator whose interface makes the weighted fit the default; documenting and testing that combination is the gap this paper fills.

2.4 Evaluating imputed files

Because an imputation is a draw from an estimated conditional distribution, pointwise accuracy metrics understate quality; the relevant question is distributional. We report pinball loss, from the quantile-loss tradition (Koenker, 2005; Meinshausen, 2006), and Wasserstein distance; as computed here both score marginal fit (Section 3.4). For the joint, the evaluation perspective this paper formalizes — one latent population, surveys as partial views of it, candidates scored through each view against held-out survey samples — has ancestry in several literatures that have not, to our knowledge, been assembled into an operational benchmark: the survey-statistics treatment of multiple sources observing one population (Lohr and Raghunathan, 2017), the missing-data view of fusion as block missingness on a single file (Little and Rubin, 2002; Rässler, 2002), strictly proper scoring of distributional claims against realized samples (Gneiting and Raftery, 2007; Székely and Rizzo, 2013), classifier two-sample tests (Lopez-Paz and Oquab, 2017), and the precision–recall–density–coverage geometry of generative-model evaluation (Naeem et al., 2020). Population-synthesis practice validates against input marginals rather than held-out views. Section 3.3 operationalizes the combination; the reweighting fragility diagnostic of Section 3.4 is, to our knowledge, new.

3 Methodology

3.1 The estimator

`microcosm-fit` models $P(\text{targets} \mid \text{predictors})$ with a quantile-regression-forest estimator (Meinshausen, 2006; Johnson, 2024) wrapped in three mechanisms.

Weighted bootstrap. Random forests cannot honor a per-record `sample_weight` in their predictive distribution: a fully grown leaf holds individual training rows, and weighting the impurity criterion does not change which values a quantile query reads out. `microcosm-fit` therefore materializes weights into the data: before each forest is grown, training rows are resampled with replacement with probability proportional to weight, so leaf distributions reflect the weighted population. Draws from the fitted forest then sample the *weighted* conditional. Resampling to undo an unequal-probability design is an established device (Hinkins et al., 1997; Dong et al., 2014); Section 2.3 places this one in that literature.

Regime gates. Economic variables are rarely smooth. Most adults hold zero dividend income; net worth is negative for some households, exactly zero for others, and spans several

orders of magnitude when positive. A single regressor fit across such a target either drops a tail or interpolates across the empty gap at zero; hurdle-style two-part models are the classical remedy (Mullahy, 1986; Lambert, 1992). `microcosm-fit` generalizes the two-part idea to the target's *sign support* — whichever of $\{\text{negative}, \text{zero}, \text{positive}\}$ actually occur — and imputes in four steps:

1. *Detect the regime.* Read which sign classes occur in the training data. Detection is deliberately unweighted: which signs *exist* is a fact about the variable, not about the sampling design, and a rare but real class (negative net worth, say) must not vanish because its weight share is small.

2. *Fit the gate.* A classifier learns the probability of each sign class given the predictors. The gate is weighted directly through its `sample_weight` argument rather than by resampling the training rows — resampling could delete a rare sign class outright, exactly the failure step 1 exists to prevent.

3. *Fit magnitudes within each sign.* A separate quantile-regression forest models the magnitude of each nonzero sign class, so no forest ever straddles the gap at zero.

4. *Draw.* For each receiver row, sample a sign class from the gate's predicted probabilities — not the modal class, which would collapse the zero mass — and, if the class is nonzero, draw a magnitude from that class's forest.

For dividend income the regime is $\{0, +\}$ and the gate is a zero hurdle; for net worth it is $\{-, 0, +\}$ and the gate is three-way. The gate's probabilities are those of the classifier as fitted on the training rows: the only behavior at the version evaluated here, and still the package's default. For a rare sign class such a gate can be overconfident on rows it has not seen, and later versions of `microcosm-fit` add an opt-in out-of-fold calibration of the gate for that case (PolicyEngine, 2026). The tasks in this paper contain no rare sign class on which the methods differ, so the gate's behavior on rare classes is not evaluated here. Section 5.2 compares the gated estimator with ungated baselines on zero-inflated targets.

Sequential chaining. Targets are imputed in order, each conditioning on the predictors plus the targets already drawn (Raghunathan et al., 2001; Van Buuren and Groothuis-Oudshoorn, 2011), so dependence among imputed variables survives.

Interface contract. Fitting over a MICROCOSM frame reads the frame's typed survey weights by default; fitting over a plain data frame — the standalone path — requires the

Method	Weights	Gates	Chaining	Implementation
`microcosm-fit`	design	yes	yes	`microcosm-fit`
– unweighted	none	yes	yes	`weights="none"`
– unchained	design	yes	no	targets fit independently
– ungated	design	no	yes	single-sign path (see text)
– ungated, unchained	design	no	no	both of the above
– net worth derived	design	yes	yes	net worth computed from three imputed components
– unchained, net worth derived	design	yes	no	the same, components fit independently
QRF (`microimpute`)	design	no	yes	`microimpute` QRF, as shipped
OLS	design	no	no	`microimpute`
Linear quantile regression	design	no	no	`microimpute`
NND hot deck	standardization only	via donation		`py-statmatch`, nearest donor
Weighted random hot deck	standardization, donor choice	via donation		`py-statmatch`, RANDwNND
Constrained hot deck	standardization only	via donation		`py-statmatch`, capped donor use
Weighted marginal draw	design	—	—	unconditional weighted draw

Table 1: The method surface. Every fit that has a weights column uses the survey's design weights; the one exception is the unweighted ablation, whose sole difference from the full estimator is `weights="none"`. The four ablations are the full estimator with one or two design choices knocked out and nothing else changed. The two net-worth-derived variants run only in the production-scale profile, where the targets form a balance sheet. The `microimpute` forest is an external baseline run as shipped: it keeps the design weights and chains its targets, but it has no gate and it draws through its own sampler, so it is not a knockout of the full estimator. The hot decks donate all target values from one matched donor record, which preserves zero masses and cross-target structure by construction rather than by modeling them; they differ in which donor is taken, and only the weighted random hot deck uses the donor weights for that choice.

weights explicitly (a weight column name, a weight vector, or the literal `"none"`), and omitting them is an error rather than a silent unweighted fit. The contract makes the weight-blindness failure mode of Section 2.3 unrepresentable by accident.

3.2 Ablations and baselines

Table 1 lays out the method surface. Each design choice is evaluated by knocking it out with everything else held fixed: the full estimator versus itself with `weights="none"` (the weighted bootstrap's contribution), versus fitting each target independently (chaining's contribution), and versus itself with the gates off (gating's contribution). `microcosm-fit` exposes no gates-off switch, but it builds a gate only for a target whose training values fall in more than one sign class; a strictly positive target is fitted with one forest and no gate. The gates-off ablation uses that path. Each target with more than one sign class is shifted by a constant that makes all its training values positive, the estimator is fitted as usual, and the constant is subtracted from the draws. The weighted bootstrap, the forests, the draw, and the chaining are the estimator's own; the one difference is that a single forest covers all rows, zeros included, where the full estimator has a classifier and a forest per sign. A

target that already has one sign class is not shifted, and there the ablation is the full estimator. Knocking out gates and chaining together completes a two-by-two design.

Where the targets satisfy an accounting identity, a practitioner would impute all but one of them and compute the last. The net-worth-derived variants do so in the production-scale profile: they impute the three balance-sheet components, chained or independently, and set net worth to their signed sum. They separate what chaining contributes through the identity from what it contributes through the dependence among the components.

Baselines are standard practice: quantile-regression-forest, ordinary-least-squares, and linear quantile-regression imputers from `microimpute` (PolicyEngine, 2025), and distance hot-deck statistical matching from `py-statmatch`, a Python port of the R package StatMatch (D'Orazio, 2025). The `microimpute` forest is built on the same forest library as the full estimator (Johnson, 2024), but it is not a knockout of it. It chains its targets, and it turns each requested quantile level into a draw from a ten-level conditional quantile grid (levels $1/11$ to $10/11$) at a randomly chosen index. Under the level sampling used for all baselines here, that draws the lowest and highest levels 25 and 21 percent of the time, against 10 percent each for a uniform draw. Its random state is also fixed.

The three hot decks share one distance: Euclidean on the matching variables, which `py-statmatch` standardizes by weighted moments when the donor weights are unequal. They differ in the donor taken. The nearest-neighbour hot deck takes the closest donor, so the weights do not enter its choice. The weighted random hot deck draws one of the $\lceil \sqrt{n} \rceil$ nearest of the n donors with probability proportional to its weight (D'Orazio et al., 2006). The constrained hot deck finds the assignment of least total distance in which no donor serves more than $\lceil n_{\text{receivers}}/n \rceil$ receivers; the cap counts receivers and does not use the weights. Its assignment problem is dense, and at the population view's 20,000 receivers one split took thirteen minutes against seconds for every other method, so it is run on the within-survey tasks only.

3.3 The population-view harness

Per-variable metrics cannot adjudicate the question this paper cares about, because the methods differ most in how they treat the *joint*. We therefore evaluate within a framework we call the population-view harness, sketched in Figure 1.

The framework has three ingredients. One latent population distribution P produces individuals. Each survey s observes it through a *view* V_s — a variable subset, a sampling design,

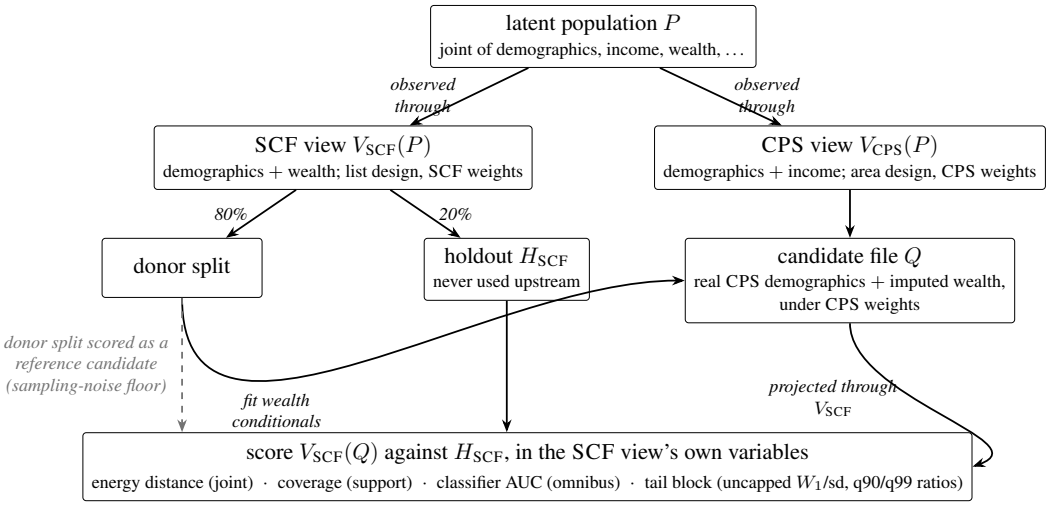


Figure 1: The population-view harness. One latent population is observed through survey views — here the SCF (demographics and wealth, list design) and the CPS (demographics and income, area design). A candidate file assembles real CPS demographics with imputed wealth; the harness projects it through the SCF view and scores it against a held-out SCF sample on four blocks. The holdout appears nowhere upstream of any candidate, and the donor split itself is scored as a reference candidate to calibrate sampling noise.

a measurement idiom — and the survey's holdout H_s is a weighted sample from $V_s(P)$ (Lohr and Raghunathan, 2017). A candidate weighted file Q (any generator's) is scored by projecting it through each view and comparing $V_s(Q)$ to H_s *in the view's own variable space*. Cross-survey consistency is never required — the candidate is only asked to explain each view in its own idiom — which keeps the harness usable even though surveys disagree with one another about the population they share.

Each view carries four complementary blocks, chosen so that no single failure mode can hide. For weighted samples $X = \{(x_i, a_i)\}$ from the candidate and $Y = \{(y_j, b_j)\}$ from the holdout (weights normalized to sum to one):

Energy distance (joint fit; lower is better, 0 is perfect). The weighted energy distance (Székely and Rizzo, 2013)

$$\mathcal{E}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\| - \sum_{i,i'} a_i a_{i'} \|x_i - x_{i'}\| - \sum_{j,j'} b_j b_{j'} \|y_j - y_{j'}\|$$

is the sample form of a strictly proper scoring rule (Gneiting and Raftery, 2007): its expectation is uniquely minimized by the true distribution, so a candidate hedged toward modal households scores strictly worse than one matching the full joint.

Coverage (support; higher is better, 1 is perfect). PRDC coverage (Naeem et al., 2020) is the weighted fraction of holdout points with at least one candidate point inside their k-nearest-neighbour radius — does the candidate reach every neighbourhood the holdout occupies? It is the explicit anti-collapse axis, and because it depends on the candidate only through its *support*, it is by definition invariant to any reweighting of the candidate: the one block a later calibration cannot change. As implemented, each sample larger than 2,048 points is first reduced to that size by a weight-proportional resample (the cap the energy distance shares). The candidate always is, so the reported coverage is invariant to reweighting only up to that resample.

Classifier two-sample test (omnibus; 0.5 is indistinguishable). A gradient-boosted classifier is trained to tell candidate rows from holdout rows at equal class mass, and we report its cross-validated AUC (Lopez-Paz and Oquab, 2017). This is the omnibus block — a single overall test sensitive to *any* difference in the joint distribution, including dependence violations that leave every marginal intact. An AUC of 0.5 means the classifier cannot separate the files; 1.0 means it always can.

Tail block (extremes; ratio of 1 is perfect). The fourth block exists because we caught the first three missing a real failure. Pairwise sample-geometry metrics run on capped subsamples, and a candidate whose imputed wealth 99th percentile is double the holdout's can still tie on energy distance: the discrepant mass is a sliver of standardized pairwise space. Economic variables live in heavy right tails, so each *imputed* column also carries a tail block computed on the full weighted samples with no cap: its weighted Wasserstein-1 distance to the holdout, scaled by the holdout's weighted standard deviation, and its weighted 90th- and 99th-percentile ratios (candidate over holdout).

Holdouts are used nowhere upstream, so the harness is a non-self-referential test surface. A sampling-noise floor — the donor split itself scored as a candidate — anchors what "as good as another sample of the survey" means on each axis, including how much a tail ratio wobbles under sampling noise alone.

3.4 Marginal and stress metrics

All metrics are computed on held-out receiver rows under survey weights, with one exception noted under pinball loss.

Pinball loss. Pinball loss — also called quantile loss; the terms are interchangeable and we use *pinball* throughout — scores predicted quantiles against realized values (Koenker, 2005). For a predicted quantile $\hat{q}_\tau$ and realized value y, the loss is $\rho_\tau(y - \hat{q}_\tau)$ with $\rho_\tau(u) = u\,(\tau - \mathbf{1}[u < 0])$; we average it, weighted, over the decile grid $\tau \in \{0.1, \ldots, 0.9\}$. A single stochastic draw carries no conditional quantiles, so $\hat{q}_\tau$ here is the τ-quantile of the method's pooled receiver draw, read with every receiver row counted once, and it is scored against every held-out value under the receiver weights. As computed, pinball loss is therefore a *marginal* score. It does not test conditional calibration and cannot separate a conditional method from an unconditional one. Because the predicted quantiles are read without weights, it also favors a draw whose unweighted quantiles already match the weighted population.

Wasserstein-1 distance. Marginal fit is scored by the weighted Wasserstein-1 distance to the weighted donor distribution,

$$W_1(F, G) = \int_0^1 \left|F^{-1}(u) - G^{-1}(u)\right| du,$$

the average absolute gap between the two distributions' quantile functions — equivalently, the area between their CDFs. The subscript is the order: W_1 averages absolute quantile gaps in the variable's own units (dollars here), where W_2 would square them. We use order 1 because it is interpretable in dollars and does not let a single extreme quantile dominate the score quadratically.

Zero-share error. The absolute difference between the candidate's and the donor's weighted share of exact zeros — preservation of the zero mass of zero-inflated targets.

Reweighting fragility. Fragility formalizes the landmine diagnostic of Section 1: over the family of bounded multiplicative reweightings $w_i \mapsto m_i w_i$ with $m_i \in [1/\kappa, \kappa]$ (the hard weight-ratio bounds production calibration guards use), the worst-case single-record share of a target aggregate has the closed form $\kappa^2 c^* / (\kappa^2 c^* + S - c^*)$, where c^* is the largest baseline contribution $w_i|a_i|$ and S their sum. Read it as: after the most adversarial reweighting a calibrator could apply, what share of the aggregate can one record carry? We report it at

$\kappa = 5$ for every method next to the held-out truth's own fragility — the exposure a faithful method should *match*, not minimize.

3.5 Protocol

Every task uses an 80/20 donor/receiver holdout repeated over ten seeds, with splits paired across methods: the split is a pure function of the seed, so every method fits and is scored on identical partitions. Task inputs are frozen once and pinned by hash; only the method varies within a sweep. All results tables are generated from the sweep artifacts by the repository's command line; no result is hand-entered. Where a method is unavailable or a metric undefined for a task, the cell is reported as absent rather than omitted.

4 Data and tasks

Four task families span the settings the estimator is used in. Where a task imputes within one survey, the protocol of Section 3.5 applies: 80/20 donor/receiver holdouts repeated over ten paired seeds. The population-view experiment additionally projects candidates onto a second survey. Every table in Section 5 regenerates from the repository's committed run configurations; row caps and skipped cells are recorded in the run manifests.

4.1 Task 1: wealth within the SCF

The Survey of Consumer Finances (Board of Governors of the Federal Reserve System, 2023) observes household wealth in detail. From the 2022 summary extract (first implicate, survey weights) we impute `debt` (zero-inflated and nonnegative) and `networth` (sign-mixed, with a heavy right tail) from age, sex, education class, marital status, children, total income, and wage income. The two targets are chosen deliberately: their sign structures exercise the regime gates, and their strong mutual dependence exercises chaining. Appendix A reruns this task and both profiles of the next on the 2025 wave, which no method here was tuned on.

4.2 Task 2: the SCF→CPS population view

Each method fits wealth conditionals on an SCF donor split and imputes them onto real CPS ASEC households sharing the predictor set — households built from the Census Bureau's ASEC public-use files (U.S. Census Bureau, 2025), read directly from `census.gov` so that no task input flows through any PolicyEngine processing. The experiment runs in two profiles. The *minimal* profile — six shared predictors, the two wealth targets of Section 4.1,

a single receiver vintage (ASEC 2025) — is the controlled instantiation in which every mechanism is legible. The *production-scale* profile mirrors the production stack's shape in three ways. The shared predictor set widens to everything the two surveys genuinely share: ten variables, adding education class, race, homeownership, and labor-force status, each mapping verified against both files' codebooks. The receiver pools three ASEC vintages (2023–2025, weights scaled to one household population, 168,852 households), as the production support spine does. And the targets widen to a four-component chained balance sheet — financial assets, nonfinancial assets, debt, then net worth — whose accounting identity networth $=$ fin $+$ nfin $-$ debt holds exactly in the extract, so chaining's contribution is directly interpretable. In both profiles each run imputes onto a fixed uniform subsample of 20,000 receiver households (of 55,762 in the minimal profile and 168,852 at production scale), the same for every method and seed. The candidate population — CPS demographics plus imputed wealth, under CPS household weights — is then scored through the harness of Section 3.3 against the *held-out* SCF: the joint of shared predictors and wealth, under SCF weights. The `scf_sample_reference` rows score the donor split itself against the holdout, anchoring the sampling-noise floor. This is the paper's direct test of whether conditional-distribution methods preserve joint structure that matching's conditional-independence gluing loses (Rässler, 2002; D'Orazio et al., 2006), measured where it matters: on a receiver file the donor survey never saw. Because the demographic block of every candidate is the same real CPS sample, differences between methods isolate the wealth block and its coupling to demographics; the shared demographic offset between the two surveys is common to all methods and bounded below by the reference floor.

4.3 Task 3: zero-inflated income components in the CPS

From the Census Bureau's ASEC 2025 public-use person file (income reference year 2024), read directly from `census.gov` — never a processed or enhanced artifact, which would embed the production imputations this paper is about — we impute interest and dividend income for adults from age, sex, and employment income, under the ASEC person supplement weights. (Survey files publish per-person design weights; MICROCOSM's production convention deliberately carries calibrated *household* weights onto persons, since its calibration operates at household grain. The task fixes the survey's own published weights, and because every method shares whatever weights the task fixes, the method comparison does not turn on this choice.) Interest is zero for 41.7 percent of adults and dividends for 84.7 percent (weighted shares), both with heavy positive tails — the regime surface the gates are built for.

The task runs on a fixed uniform subsample of 20,000 of the file's 108,044 adults. Public-use ASEC amounts are topcoded, which truncates the extreme tail relative to an administrative or SCF-style source; the fragility comparisons of Section 5.2 are therefore conservative. This task also carries the reweighting stress test of Section 3.4. The test is motivated by prior POLICYENGINE production experience, in which an unweighted fit broadcast rare high-value donor records across a receiver file as near-zero-weight point masses that later reweighting amplified. All fragility numbers reported here are measured under this paper's protocol.

4.4 Task 4: cross-dataset transfer

To test behavior beyond economic microdata, we run the method surface over the six OpenML AutoML Benchmark regression datasets used in the `microimpute` manuscript's appendix (Vanschoren et al., 2014) — `space_ga`, `elevators`, `brazilian_houses`, `onlinenewspopularity`, `abalone`, and `house_sales` — treating each dataset's target as the imputed column under uniform weights. The two datasets with more than 20,000 rows, `house_sales` and `onlinenewspopularity`, are capped at 20,000 by a fixed uniform subsample.

5 Results

Throughout, *the full estimator* names `microcosm-fit` with all three design choices on (weighted, gated, chained); each table indents its ablations and variants beneath it. The `microimpute` forest, OLS, quantile regression, and the hot decks are the standard-practice baselines (Table 1), and the weighted marginal draw is the unconditional lower bound. A row appears only where its method applies: the net-worth-derived variants at production scale, the constrained hot deck on the within-survey tasks. Unless a caption says otherwise, every cell is the mean over the ten paired seeds with the standard deviation across seeds in parentheses, and bold marks the best non-reference value in a column: lowest for distances and losses, highest for coverage, closest to 0.5 for the two-sample AUC, closest to 1 for tail ratios, and closest to the held-out truth's own value for fragility.

5.1 Wealth within the SCF

Table 2 reports the within-SCF task. The headline is the weighting ablation: switching the full estimator to `weights="none"` degrades net-worth Wasserstein-1 by a factor of six

Method	Debt pinball	Debt W_1	Debt zero-share err.	Net worth pinball	Net worth W_1
microcosm-fit	49,201 (2,637)	13,001 (6,285)	0.020 (0.012)	551,317 (36,012)	**135,871** (51,106)
– unweighted	50,218 (2,452)	32,247 (16,930)	0.012 (0.014)	703,457 (77,173)	829,158 (445,108)
– unchained	49,201 (2,637)	13,001 (6,285)	0.020 (0.012)	556,622 (23,349)	158,606 (44,310)
– ungated	49,149 (2,832)	12,714 (4,817)	0.072 (0.029)	563,223 (35,315)	180,233 (52,533)
– ungated, unchained	49,149 (2,832)	12,714 (4,817)	0.072 (0.029)	571,907 (35,101)	180,570 (86,541)
QRF (microimpute)	50,433 (3,120)	16,619 (5,358)	0.018 (0.018)	560,819 (36,903)	234,367 (142,057)
OLS	70,053 (17,637)	156,295 (74,822)	0.226 (0.018)	1,116,572 (383,007)	3,565,923 (2,062,063)
Quantile regression	49,797 (3,391)	29,460 (20,456)	0.151 (0.031)	534,420 (37,801)	295,470 (110,986)
NND hot deck (py-statmatch)	49,537 (2,627)	13,770 (5,141)	0.017 (0.013)	644,921 (51,255)	509,013 (399,757)
Weighted random hot deck	48,802 (2,595)	**11,395** (4,789)	0.020 (0.014)	506,558 (29,427)	172,984 (48,701)
Constrained hot deck	49,518 (2,706)	16,244 (6,243)	**0.011** (0.007)	642,310 (44,183)	642,051 (529,395)
Weighted marginal draw	**48,277** (2,586)	14,617 (4,732)	0.020 (0.017)	**443,554** (20,651)	215,398 (87,264)

Table 2: Wealth imputation within the SCF (debt and net worth from demographics and income): weighted pinball loss, Wasserstein-1 to the weighted donor distribution, and zero-share error, mean (sd) over ten paired seeds. Bold marks the best non-reference value per column. Lower is better.

(135,871 to 829,158) and debt by a factor of 2.5 (13,001 to 32,247). The SCF's list-sample design oversamples wealthy households at low weight, so an unweighted fit learns the sample's wealth distribution rather than the population's — this is the weight-blindness failure mode of Section 1 measured directly, and no other design choice comes close to its effect size. Removing the gates raises net-worth W_1 by a third (to 180,233) and removing chaining by 17 percent (to 158,606); both paired deltas are smaller than their own standard deviation across seeds (Table 5). What the gates change here is the zero mass: with them off, debt zero-share error rises from 2.0 to 7.2 percentage points.

Against matching, the comparison turns on how the donor is chosen. The nearest-neighbour hot deck nearly ties the full estimator on debt marginal fit (13,770 vs. 13,001) but is 3.7 times worse on net worth (509,013). The weighted random hot deck, which draws among the nearest donors in proportion to their weights, is not: it posts the lowest debt W_1 of any method (11,395) and 172,984 on net worth, 1.3 times the full estimator and inside its seed-to-seed spread. The constrained hot deck, whose cap counts receivers and ignores the weights, does no better than the nearest-neighbour one (642,051). Within the survey, then, the full estimator's large margin over nearest-neighbour matching is not a margin over matching as such: the one hot deck whose choice of donor uses the weights closes most of it, and the two whose choice ignores them do not. Among the other baselines, the `microimpute` forest sits at 234,367 on net worth, and OLS is catastrophic on both targets (156,295 and 3.6 million): normal residuals cannot represent a wealth distribution. Linear quantile regression leaves 15 percent of the debt zero mass unaccounted for, against 7 percent for the gates-off ablations and at most 2 percent for the other forest-based methods.

Method	Interest pinball	Interest W_1	Interest zero-share	Dividend pinball	Dividend W_1	Dividend zero-share
microcosm-fit	1,491 (63.58)	289.66 (110.29)	0.013 (0.011)	320.68 (45.26)	102.03 (37.36)	0.006 (0.004)
– unweighted	1,491 (63.62)	263.38 (74.44)	0.011 (0.007)	320.68 (45.29)	138.14 (56.50)	0.008 (0.005)
– unchained	1,491 (63.58)	289.66 (110.29)	0.013 (0.011)	320.69 (45.26)	113.33 (82.23)	**0.006** (0.005)
– ungated	1,491 (63.68)	278.16 (166.32)	0.024 (0.015)	321.05 (45.27)	222.61 (31.75)	0.072 (0.006)
– ungated, unchained	1,491 (63.68)	278.16 (166.32)	0.024 (0.015)	320.99 (45.26)	209.37 (53.28)	0.043 (0.009)
QRF (microimpute)	1,493 (63.34)	406.35 (173.82)	0.014 (0.011)	322.17 (45.24)	410.69 (173.69)	0.044 (0.010)
OLS	2,527 (63.57)	8,637 (250.17)	0.422 (0.012)	961.52 (112.29)	4,404 (783.34)	0.844 (0.005)
Quantile regression	1,493 (63.23)	711.73 (142.92)	0.270 (0.016)	320.91 (45.19)	275.50 (38.14)	0.251 (0.011)
NND hot deck (py-statmatch)	1,493 (63.89)	759.97 (120.26)	0.021 (0.019)	320.94 (45.22)	227.10 (49.73)	0.009 (0.009)
Weighted random hot deck	**1,491** (63.49)	232.16 (85.69)	**0.009** (0.009)	320.68 (45.28)	107.16 (29.60)	0.007 (0.008)
Constrained hot deck	1,491 (63.38)	**230.52** (82.36)	0.018 (0.013)	320.75 (45.27)	**97.88** (29.10)	0.007 (0.005)
Weighted marginal draw	1,491 (63.61)	287.37 (99.08)	0.011 (0.013)	**320.68** (45.25)	127.57 (68.67)	0.006 (0.004)

Table 3: Zero-inflated income components in the raw CPS (interest and dividend income for adults, on a fixed subsample of 20,000 of the file's 108,044 adults): weighted pinball loss, Wasserstein-1 to the weighted donor distribution, and zero-share error, mean (sd) over ten paired seeds. Bold marks the best non-reference value per column; lower is better.

One result reads wrong until the metric is examined: the *unconditional* weighted marginal draw posts the best pinball loss on both targets. Pinball loss as computed here is a marginal score whose predicted quantiles are read without receiver weights (Section 3.4). In a survey that oversamples the wealthy, a conditional method's pooled draw follows the receiver sample's composition, while the marginal draw is centered on the weighted population by construction, so this column favors the marginal draw for a reason unrelated to conditional fit, and we do not read its lead as evidence about the methods. More generally, no receiver-side marginal metric can distinguish a method that draws from the correct weighted marginal while ignoring predictors from one that also gets the conditionals right. That is what the population-view harness of Section 5.5, which scores the joint, is for.

5.2 Zero-inflated components and reweighting fragility

Table 3 reports the CPS components task. Pinball loss barely separates the non-OLS methods — as a marginal score it is not built to — but the zero mass separates them sharply. The full estimator holds zero-share error to 0.6 and 1.3 percentage points (dividends, interest). With its gates off, the same estimator misses by 7.2 and 2.4 points, and its dividend W_1 doubles (223 against 102). It errs by drawing too many zeros: in a forest grown to purity an atom at zero forms larger leaves than the positive values do, and a forest that pools leaf samples across trees over-represents the atom in each row's conditional. The gate replaces that with a classifier's probability. The `microimpute` forest, which has no gate either and also differs in its sampler (Section 3.2), misses the dividend zero share by 4.4 points; linear quantile regression misses by 25–27 points because a 99-level linear quantile grid cannot hold a five-sixths zero mass; and OLS misses the dividend zero share by 84 points — with

Method	Interest fragility	(truth)	Dividend fragility	(truth)
microcosm-fit	0.384 (0.051)	0.372 (0.043)	0.675 (0.100)	0.614 (0.104)
– unweighted	0.395 (0.043)	0.372 (0.043)	0.699 (0.057)	0.614 (0.104)
– unchained	0.384 (0.051)	0.372 (0.043)	0.661 (0.100)	0.614 (0.104)
– ungated	0.384 (0.043)	0.372 (0.043)	0.781 (0.052)	0.614 (0.104)
– ungated, unchained	0.384 (0.043)	0.372 (0.043)	0.716 (0.068)	0.614 (0.104)
QRF (microimpute)	0.364 (0.064)	0.372 (0.043)	0.633 (0.087)	0.614 (0.104)
OLS	0.055 (0.010)	0.372 (0.043)	0.055 (0.009)	0.614 (0.104)
Quantile regression	**0.368** (0.105)	0.372 (0.043)	0.444 (0.073)	0.614 (0.104)
NND hot deck (py-statmatch)	0.391 (0.027)	0.372 (0.043)	0.647 (0.140)	0.614 (0.104)
Weighted random hot deck	0.410 (0.085)	0.372 (0.043)	0.761 (0.066)	0.614 (0.104)
Constrained hot deck	0.379 (0.044)	0.372 (0.043)	**0.602** (0.131)	0.614 (0.104)
Weighted marginal draw	0.356 (0.042)	0.372 (0.043)	0.746 (0.067)	0.614 (0.104)

Table 4: Reweighting fragility at $\kappa = 5$: worst-case single-record share of each component's aggregate over bounded multiplicative reweightings, next to the held-out truth's own fragility. The target is the truth's exposure — bold marks the closest method; substantially lower means the method fails to generate realistic tails, substantially higher means it plants landmines.

normal residual draws, nearly every adult becomes a dividend recipient. The hot decks, which donate observed values, hold the zero mass within 0.7–2.1 points, as expected. On marginal fit the nearest-neighbour hot deck's W_1 is more than twice the full estimator's on both components (227 and 760 against 102 and 290), while the weighted random and constrained hot decks match or beat it (107 and 232; 98 and 231). The pattern differs from the SCF's in one respect. There the constrained hot deck, which ignores the weights, did no better than the nearest-neighbour one; here it does as well as the weighted one. With person weights that carry little outcome information, what the nearest-donor rule loses on this task it loses to donor reuse, which the cap and the random draw both limit, and not to weight-blindness.

Table 4 reports the landmine diagnostic. Two readings matter. First, the held-out truth is itself concentrated: under $\kappa = 5$ reweighting a single record can carry 37 percent of the interest aggregate and 61 percent of the dividend aggregate at this sample size — heavy-tailed capital income is genuinely fragile, and a faithful method should *match* that exposure, not minimize it. Second, every forest- and donation-based method tracks the truth's fragility closely, while OLS sits far below it, at 0.06 against 0.37 and 0.61 — understating the true exposure by factors of roughly 7 and 11. Its aggregate is spread across many moderate records because the method never generates realistic tails; the zero-share and Wasserstein numbers show the same failure from other angles. The highest dividend exposures belong to the gates-off ablation (0.78), the weighted random hot deck (0.76), and the marginal draw (0.75), against a

Task	Target	Metric	Δ unweighted	Δ unchained	Δ ungated	Δ ungated, unchained
cps components	dividend_income	pinball_loss	0.000 (0.056)	0.005 (0.033)	0.361 (0.097)	0.302 (0.089)
cps components	dividend_income	wasserstein1	36.11 (46.53)	11.29 (94.58)	120.57 (55.98)	107.34 (74.80)
cps components	dividend_income	zero_share_error	0.002 (0.006)	-0.000 (0.006)	0.067 (0.008)	0.037 (0.010)
cps components	interest_income	pinball_loss	-0.016 (0.307)	0.000 (0.000)	-0.024 (0.347)	-0.024 (0.347)
cps components	interest_income	wasserstein1	-26.28 (114.17)	0.000 (0.000)	-11.50 (187.77)	-11.50 (187.77)
cps components	interest_income	zero_share_error	-0.002 (0.006)	0.000 (0.000)	0.011 (0.023)	0.011 (0.023)
scf wealth	debt	pinball_loss	1,017 (709.88)	0.000 (0.000)	-51.77 (595.03)	-51.77 (595.03)
scf wealth	debt	wasserstein1	19,246 (16,086)	0.000 (0.000)	-287.07 (7,733)	-287.07 (7,733)
scf wealth	debt	zero_share_error	-0.008 (0.011)	0.000 (0.000)	0.052 (0.034)	0.052 (0.034)
scf wealth	networth	pinball_loss	152,140 (51,569)	5,305 (23,838)	11,906 (23,086)	20,590 (19,445)
scf wealth	networth	wasserstein1	693,287 (431,527)	22,735 (70,684)	44,362 (67,012)	44,699 (93,246)
scf wealth	networth	zero_share_error	-0.000 (0.001)	0.000 (0.001)	0.001 (0.003)	0.001 (0.002)

Table 5: Paired-seed deltas from the full estimator, mean (sd) over ten seeds, for its four ablations: weights off, chaining off, gates off, and gates and chaining off. Positive means the ablation is worse on these lower-is-better metrics; a delta near zero means the knocked-out design choice is inert on that task. On each task's first target (interest, debt) there is nothing to chain on, so the chaining column is zero and the two gates-off columns agree.

truth of 0.61 whose own standard deviation across seeds is 0.10; none approaches a landmine on this within-survey task. The landmine pathology of Section 1 arises when unweighted cross-file fits meet downstream reweighting, and the direct evidence for it here is the tail block of the population-view experiment (Section 5.5).

5.3 Ablations: attributing the gains

Table 5 attributes the full estimator's behavior to its design choices. Each cell answers one question: on the same ten splits, how much did this metric worsen when one design choice was knocked out? The $+693{,}287$ under Δ unweighted on net-worth W_1, for instance, reads: refitting the identical estimator without weights raised the marginal-fit distance by \$693,287 on average. Because ablation and estimator share each split, the delta subtracts out split-to-split noise; the parenthetical standard deviation says how much the delta itself varies across the ten splits.

The ordering is task-dependent in an informative way. On the SCF — informative weights, heavy tails — weighting dominates ($+693{,}287$ net-worth W_1 when removed). Gating ($+44{,}362$, sd $67{,}012$) and chaining ($+22{,}735$, sd $70{,}684$) shift net-worth W_1 by less than their dispersion across seeds, and gating's clear effect is on the debt zero share ($+0.052$, sd 0.034). On the CPS components — weakly informative weights, extreme zero inflation — weighting has a small effect (removing it raises dividend W_1 by 36, sd 47, from 102, and leaves the other metrics within seed noise), chaining is inert, and gating carries the load: with the gates off, dividend zero-share error rises by 6.7 percentage points (sd 0.8) and dividend W_1 by 121 (sd 56). Interest, whose zero share is nearer one half, moves less (1.1 points, sd

Method	abalone	brazilian_houses	elevators	house_sales	onlinenewspopularity	space_ga	Mean rank
microcosm-fit	0.254	166.64	0.000	7,080	274.67	0.008	4.42
– unweighted	0.257	184.92	0.000	6,744	233.10	0.008	4.00
– ungated	0.254	166.64	0.000	7,080	274.67	0.009	4.58
QRF (microimpute)	0.174	191.36	0.000	8,461	546.16	0.012	5.67
OLS	0.532	212.09	0.002	90,518	6,117	0.015	9.83
Quantile regression	0.321	212.04	0.001	24,115	382.98	0.085	9.00
NND hot deck (py-statmatch)	0.176	199.12	0.000	10,212	161.43	0.010	4.67
Weighted random hot deck	0.237	196.93	0.000	16,570	217.64	0.010	5.33
Constrained hot deck	0.151	198.10	0.000	9,248	144.90	0.009	**3.50**
Weighted marginal draw	0.148	210.50	0.000	7,135	143.52	0.011	4.00

Table 6: Wasserstein-1 to the donor distribution across the six OpenML regression datasets (uniform weights), mean over ten paired seeds, with mean rank across datasets among the ten methods shown. The two unchained arms are omitted: with a single target they coincide with their chained counterparts. The two datasets with more than 20,000 rows (`house_sales`, 21,613, and `onlinenewspopularity`, 39,644) are capped at 20,000 by a fixed uniform subsample.

2.3). Which choice binds depends on the survey's design and the target's structure. Where a choice does not bind, its cost is small: the unweighted ablation is slightly ahead on SCF debt zero share (0.012 against 0.020, Table 2) and on the uniform-weight OpenML suite (Section 5.4).

5.4 Cross-dataset transfer

Table 6 runs the surface over the six OpenML regression datasets under uniform weights. Two observations. First, with weights uninformative by construction, the full estimator and its unweighted ablation differ only by a uniform resample of the training rows, and they rank in the upper half (mean ranks 4.4 and 4.0) alongside the unconditional marginal draw (4.0); the linear methods rank last (9.0 and 9.8). The gates-off ablation equals the full estimator on five of the six datasets, whose targets have a single sign class and so no gate to remove. Second — and more important for this paper's argument — a method with no conditional structure at all ties for second of the ten on a *marginal* distance, ahead of the full estimator, and the best mean rank (3.5) goes to the constrained hot deck, which here uses each donor at most once and so reproduces the donor marginal almost by construction. Marginal metrics saturate: they reward drawing from the right marginal and cannot rank what imputation is actually for. The population-view harness exists because of exactly this ceiling.

5.5 The SCF→CPS population view

Tables 7 and 8 report the population-view experiment in its two profiles. The floor behaves as designed in both: another sample of the same survey is statistically indistinguishable

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.009 (0.006)	0.987 (0.009)	0.496 (0.023)	0.013 (0.003)	1.10 (0.202)
microcosm-fit	0.139 (0.026)	0.960 (0.014)	0.732 (0.013)	0.031 (0.007)	0.696 (0.112)
– unweighted	0.140 (0.026)	0.958 (0.014)	**0.731** (0.016)	0.159 (0.041)	1.93 (0.255)
– unchained	0.139 (0.026)	0.958 (0.016)	0.742 (0.013)	0.029 (0.009)	0.765 (0.177)
– ungated	**0.138** (0.027)	0.955 (0.016)	0.737 (0.014)	0.027 (0.006)	0.847 (0.186)
– ungated, unchained	0.138 (0.026)	0.954 (0.016)	0.747 (0.016)	0.036 (0.023)	0.822 (0.207)
QRF (microimpute)	0.140 (0.027)	0.957 (0.012)	0.775 (0.008)	0.029 (0.007)	0.805 (0.156)
OLS	0.166 (0.045)	0.732 (0.087)	0.988 (0.005)	0.434 (0.282)	**0.967** (0.505)
Quantile regression	0.141 (0.027)	0.938 (0.028)	0.927 (0.018)	0.052 (0.013)	0.530 (0.118)
NND hot deck (py-statmatch)	0.141 (0.027)	**0.962** (0.013)	0.753 (0.014)	0.051 (0.016)	0.762 (0.148)
Weighted random hot deck	0.140 (0.027)	0.958 (0.010)	0.761 (0.013)	0.050 (0.010)	0.521 (0.088)
Weighted marginal draw	0.139 (0.026)	0.937 (0.005)	0.848 (0.008)	**0.014** (0.004)	1.09 (0.222)

Table 7: The population-view experiment, *minimal* profile (six shared predictors, two targets, a fixed subsample of 20,000 of the 55,762 ASEC 2025 receiver households): candidates assembled from real CPS demographics plus each method's imputed wealth, under CPS household weights, scored against the held-out SCF through the SCF view. The first row is the sampling-noise floor; cells are mean (sd) over the ten paired donor splits. Energy distance lower is better; coverage higher; C2ST AUC of 0.5 is indistinguishable; net-worth W_1/sd lower is better; a q99 ratio of 1 is a perfect extreme tail. Bold marks the best non-reference value per column under those conventions.

(energy 0.009 and 0.012, coverage 0.987 and 0.986, AUC 0.496 and 0.499), and its q99 ratio of 1.10 calibrates how much a 99th-percentile ratio moves under sampling noise alone. Four findings follow.

First, *the weighting failure is a tail failure, and only the tail block sees it*. The unweighted ablation ties the full estimator on energy distance, coverage, and the C2ST in both profiles — and its imputed net-worth q99 is 1.93 times the holdout's in the minimal profile and 2.25 times at production scale, with tail W_1/sd five times the full estimator's (0.159 vs. 0.031 minimal; 0.133 vs. 0.027 at scale). Fitting the SCF unweighted learns the wealthy oversample's conditionals, and the resulting inflation concentrates beyond the resolution of capped pairwise geometry: we report this as a finding about evaluation practice. Any harness whose joint metrics subsample — which at these sample sizes they must — needs an uncapped tail block for heavy-tailed economic variables, or it will certify files whose top percentile is wrong by a factor of two.

Second, *chaining's contribution appears only where the targets are jointly constrained, only a joint metric sees it, and most of it is the constraint itself.* In the minimal profile (two targets) the unchained ablation is indistinguishable from the full estimator (AUC 0.742 against 0.732, under half the floor's seed-to-seed standard deviation). At production scale, where the four targets satisfy the accounting identity networth = fin + nfin − debt exactly in the donor,

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.012 (0.005)	0.986 (0.003)	0.499 (0.021)	0.013 (0.003)	1.10 (0.202)
microcosm-fit	0.118 (0.024)	0.969 (0.009)	0.761 (0.026)	0.027 (0.008)	0.767 (0.190)
– unweighted	0.120 (0.024)	**0.971** (0.008)	0.754 (0.011)	0.133 (0.038)	2.25 (0.443)
– unchained	0.118 (0.024)	0.967 (0.011)	0.819 (0.015)	0.028 (0.006)	0.699 (0.112)
– ungated	**0.117** (0.024)	0.970 (0.011)	0.758 (0.013)	0.028 (0.007)	0.767 (0.157)
– ungated, unchained	0.117 (0.024)	0.967 (0.012)	0.831 (0.012)	0.026 (0.008)	0.778 (0.129)
– net worth derived	0.118 (0.024)	0.970 (0.009)	**0.744** (0.011)	0.027 (0.008)	0.773 (0.186)
– unchained, net worth derived	0.118 (0.024)	0.970 (0.011)	0.756 (0.014)	0.031 (0.008)	0.764 (0.184)
QRF (microimpute)	0.126 (0.025)	0.965 (0.011)	0.817 (0.009)	0.033 (0.022)	1.15 (0.445)
OLS	0.152 (0.059)	0.877 (0.055)	0.995 (0.002)	0.435 (0.282)	**0.963** (0.503)
Quantile regression	0.120 (0.026)	0.951 (0.011)	0.970 (0.007)	0.051 (0.013)	0.485 (0.098)
NND hot deck (py-statmatch)	0.119 (0.025)	0.971 (0.010)	0.768 (0.009)	0.024 (0.008)	0.747 (0.107)
Weighted random hot deck	0.118 (0.025)	0.968 (0.011)	0.777 (0.006)	0.037 (0.010)	0.598 (0.110)
Weighted marginal draw	0.119 (0.024)	0.931 (0.015)	0.968 (0.003)	**0.015** (0.003)	1.12 (0.241)

Table 8: The population-view experiment, *production-scale* profile (ten shared predictors, four chain-ordered balance-sheet targets, a fixed subsample of 20,000 of the 168,852 pooled ASEC 2023–2025 receiver households). Columns, cell format, and bolding as in Table 7.

fitting them independently costs 5.8 points of C2ST AUC (0.819 vs. 0.761) while leaving every marginal and tail statistic essentially unchanged — the classifier detects balance sheets that do not add up, which no marginal metric can. The net-worth-derived variants show how much of that is the identity. Imputing the three components independently and computing net worth from them scores 0.756, level with the fully chained estimator. Chaining the three components as well scores 0.744, the lowest AUC of any method. Chaining therefore adds 1.2 AUC points beyond what the identity supplies, and deriving net worth is 1.7 points better than imputing it as a fourth chained target. Gating does not register on the joint scores of either profile: every gates-off row sits within one floor standard deviation of its gated counterpart (at scale, AUC 0.758 chained and 0.831 unchained, against 0.761 and 0.819).

Third, *richer conditioning makes the classifier two-sample test — the scorecard's omnibus block — decisive*. The unconditional weighted marginal draw — at or near the top on marginal metrics throughout this paper — moves from AUC 0.848 in the minimal profile to 0.968 at production scale, while the full estimator stays within two points of the lowest AUC in each profile (0.732 against its unweighted ablation's 0.731; 0.761 against its net-worth-derived variant's 0.744). With ten shared predictors and a four-component wealth block, a classifier finds the missing demographics–wealth coupling almost perfectly. This is the empirical case for carrying strong auxiliary predictors in a production imputation stack: richer conditioning is what turns joint-structure failures from undetectable into obvious.

Fourth, *hot-deck matching is a near-peer on this task, and the weighted random hot deck does not improve on it here*. The nearest-neighbour hot deck sits within noise of the full

estimator on energy and coverage in both profiles, within 2.1 and 0.7 AUC points, and its donated values slightly beat the full estimator's on tail W_1 at scale (0.024 vs. 0.027). The weighted random hot deck, which closed the gap within the SCF, does not carry that over: its AUC is the higher of the two hot decks' in both profiles (0.761 and 0.777, against 0.753 and 0.768), and its tail fit is worse at scale (0.037). What the full estimator adds over matching is therefore not within-survey marginal fit, once the donor is chosen with the weights (Table 2), and not the zero mass, which donation preserves (Table 3). It is a modest margin on the joint. In the minimal profile its AUC is 2.1 and 2.9 points below the two hot decks' (0.732 against 0.753 and 0.761). At scale it ties the nearest-neighbour hot deck as specified, and sits 2.4 points below it once net worth is derived (0.744 against 0.768), so that both files' balance sheets add up. Each of these margins is of the order of one floor standard deviation (0.023 and 0.021).

One more tail observation cuts against the full estimator: its imputed net-worth q99 falls short of the holdout's, at a ratio of 0.70 in the minimal profile and 0.77 at production scale. The nearest-neighbour hot deck, which donates observed values, shows the same shortfall (0.76 and 0.75), and the weighted random hot deck falls shorter still (0.52 and 0.60), while the unconditional marginal draw, which draws from the donor's weighted marginal without conditioning, sits near 1 (1.09 and 1.12, against a floor of 1.10). The shortfall is therefore shared by a forest and two donation methods that all condition on the receiver's demographics, and this experiment does not isolate its cause. No method in this surface gets the extreme tail right for free; the tail block is what keeps that visible.

The arms that distinguish these findings — the gates-off ablations, the net-worth-derived variants, and the weighted random hot deck — were also run on the 2025 wave of the SCF (Tables 12–14), outside the pre-registered replication of Appendix A. Each pattern repeats there: the debt zero share drifts with the gates off, the weighted random hot deck closes most of the nearest-neighbour hot deck's within-survey gap on net worth, and at production scale the identity recovers the unchained loss while chaining adds about a point more.

6 Discussion

The results attribute the estimator's behavior to its design choices with a regularity worth stating plainly. *Weighting is the dominant choice wherever the design is informative*: on the SCF, whose list sample oversamples wealthy households, removing the weighted bootstrap costs a factor of six on net-worth marginal fit within the donor survey and inflates the

imputed 99th percentile on the population view by a factor of 1.9 to 2.3. That effect is larger than the gap between the full estimator and every competing method except OLS. On the CPS components task, whose person weights carry little outcome information, weighting matters far less: removing it raises dividend W_1 by about a third and leaves the other metrics within seed noise. An estimator cannot know in advance which regime it is in, which is the argument for weighting by construction rather than by option. *Gating carries the zero mass*: with its gates off, the same estimator misses the dividend zero share by 7.2 percentage points instead of 0.6, and the SCF debt zero share by 7.2 instead of 2.0, because a forest that pools leaf samples over-draws an atom at zero. On the population view's joint scores every gates-off row sits within one floor standard deviation of its gated counterpart. *Chaining matters only where the targets are jointly constrained, and there most of its effect is the constraint*: invisible with two loosely coupled targets, fitting four balance-sheet components independently costs 5.8 AUC points, but computing net worth from three independently imputed components recovers that, and chaining those three adds 1.2 points more. Only the omnibus classifier sees any of it, because a balance sheet that does not add up has perfectly reasonable marginals. The practical reading is that an identity among targets should be imposed rather than learned. Where a choice does not bind, its cost is small: the unweighted ablation is slightly ahead on SCF debt zero share and on the uniform-weight OpenML suite.

Against matching, what the estimator adds is on the joint, and it is modest. A hot deck that draws its donor among the nearest in proportion to weight equals the full estimator on within-survey marginal fit, where the nearest-neighbour hot deck is 3.7 times worse on SCF net worth. That gap belongs to the nearest-donor rule, not to matching. On the SCF, whose weights are informative, the hot deck that uses them to choose the donor closes it and the constrained hot deck, which does not, leaves it open; on the CPS, whose weights carry little outcome information, both close it. All three hot decks preserve the zero mass by donation. On the population view the order reverses: the weighted random hot deck is the weaker of the two on the classifier score and on the extreme tail, and the full estimator's AUC is the lowest of the three in the minimal profile (by 2.1 and 2.9 points) and, with net worth derived, at production scale (by 2.4 and 3.3). Those margins are one to one and a half times the floor's seed-to-seed standard deviation.

The evaluation findings are as load-bearing as the method findings, and one of them corrected this paper's own harness. Marginal metrics saturate: with weak predictors, an unconditional weighted draw from the donor marginal sits at or near the top on pinball loss and Wasserstein distance — on the OpenML suite it ranks second of the eight methods — while carrying

no conditional structure at all; at production scale the classifier separates it almost perfectly (AUC 0.968 against 0.761). But the joint metrics have their own blind spot: capped pairwise geometry tied the unweighted ablation with the full estimator while its imputed net-worth 99th percentile was double the holdout's, because the discrepant mass is a sliver of standardized pairwise space. We caught this by comparing weighted quantiles directly, added the uncapped tail block to the harness, and report it as a general lesson: sample-geometry evaluations of heavy-tailed economic microdata need an explicit tail axis, with a sampling floor to calibrate its noise (the floor's own q99 ratio wobbles to 1.10). The fragility diagnostic adds the complementary caution: the target is the truth's own tail exposure, not minimal exposure, and the one method that minimizes it (OLS, at factors of 7 to 11 below truth) does so by failing to generate realistic tails at all.

Several limitations bound the claims. First, the population-view experiment runs one view pair: the common CPS–SCF offset (differing income concepts, ASEC topcoding, and income reference years spanning 2021 to 2024 across the pooled vintages) is shared by all methods but bounds how close any candidate can come to the floor. A multi-view instantiation — one candidate scored simultaneously against CPS, SIPP (U.S. Census Bureau, 2023), and PSID (Institute for Social Research, University of Michigan, 2023) views — would tighten the identification and is the production release-gate use of the harness; we leave it to future work (PSID access requires registration), since the two-view experiment already identifies the joint-structure differences this paper is about. Second, the joint-geometry blocks still run on weight-proportional subsamples (2,048 points per side); the tail block and the C2ST, which use all rows, are the axes least affected. Third, the full estimator understates the extreme tail on the population view: its imputed net-worth q99 is 0.70 to 0.77 of the holdout's. The nearest-neighbour hot deck shows the same shortfall (0.75 to 0.76), the weighted random hot deck a larger one (0.52 to 0.60), and the unconditional marginal draw none (1.09 to 1.12), so the cause is not specific to forest draws; this experiment does not isolate it. The tail block keeps the shortfall visible, and it is a caution against reading the full estimator's wins as uniform. The OLS and quantile-regression baselines are additionally converted to samplers through a 99-level grid, truncating draws beyond the 1st and 99th conditional percentiles. Fourth, the `microimpute` forest is run as shipped. It draws each value from a ten-level conditional quantile grid, which reaches no further than the 9th and 91st conditional percentiles and over-draws those two levels, and its constructor takes no seed, so its forest and its sampler run on one fixed internal seed in every cell. Its paired-seed dispersion therefore carries no forest noise, and its contrasts with the full estimator mix

the gate with the sampler. Fifth, comparisons hold shared forest backends and default hyperparameters fixed rather than tuning each method per task; the four ablations are exact controls, the cross-family comparisons are defaults-versus-defaults. Sixth, two arms are evaluated on part of the surface only. The constrained hot deck's assignment problem is too large at the population view's 20,000 receivers, so it is absent there, and the gates are tested on targets whose sign classes are all common, not on a rare class. Finally, we evaluate single imputations distributionally and do not propagate imputation uncertainty in the multiple-imputation sense (Rubin, 1987); for the population statistics microsimulation consumes, the distributional view is the relevant one, but inference from imputed files would require the fuller treatment.

7 Conclusion

Imputation sits upstream of every population statistic a microsimulation model produces, and the imputed files are reweighted by consumers the imputer never sees. We presented an estimator that treats survey weights as a structural obligation rather than an option — weighted bootstrap into the forests, directly weighted regime gates, sequential chaining, and an interface that refuses a silent unweighted fit — and a population-view harness that evaluates any candidate population against held-out survey views with a strictly proper joint score, a support-based coverage axis, a classifier two-sample test, and an uncapped tail block.

Empirically, the interface rule is the paper's largest single effect. On a survey whose design is informative, the identical estimator fit without weights is six times worse on net-worth marginal fit, and on the population view it inflates the imputed 99th percentile by a factor of 1.9 to 2.3. Capped joint geometry certifies that failure; only the tail block catches it, which is itself the paper's central lesson about evaluating imputed files. Chaining matters where targets are jointly constrained, and there most of its effect is the constraint: fitting a balance sheet's components independently costs 5.8 AUC points at production scale, imposing the accounting identity recovers them, and chaining adds 1.2 more. The gates hold the zero mass of zero-inflated components to about one percentage point; with them off the same estimator misses the dividend zero share by 7, and linear quantile regression misses both components by 25 to 27. Matching that chooses donors in proportion to weight equals the estimator on marginal fit within a survey and trails it by 1.6 to 2.9 AUC points on the joint across surveys. And marginal metrics — the field's default — rank an unconditional marginal draw at or near the top throughout, while the harness separates it decisively. The estimator is available

standalone as `microcosm-fit`; the harness, task configurations, and every number in this paper regenerate from the accompanying repository.

Conflict of interest

All authors are affiliated with POLICYENGINE, the nonprofit organization that develops MICROCOSM, `microcosm-fit`, and `microimpute`, the software evaluated in this paper. The authors have no other competing interests.

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Data and code availability

The paper and experiment code are at https://github.com/PolicyEngine/imputation-paper; every results table regenerates from the committed run artifacts via the repository's command line. The estimator is implemented in `microcosm-fit`, part of MICROCOSM, open source at https://github.com/PolicyEngine/microcosm. It is not on PyPI. It installs on its own, without the rest of the stack, as a Git dependency on the `packages/microcosm-fit` subdirectory of that repository; the paper repository's README gives the one-line command. Baseline implementations are `microimpute` (https://github.com/PolicyEngine/microimpute) and `py-statmatch` (https://github.com/CosilicoAI/py-statmatch). The reported sweeps ran against the version of `microcosm-fit` merged in commit `b538d8a` of the MICROCOSM repository (then named `populace`; that commit ships the distribution as `populace-fit`, under `packages/populace-fit`), `microimpute` 1.15.1, and `py-statmatch` at commit `094a242`.

The inputs are public and need no registration: the SCF 2022 summary extract from the Federal Reserve (SHA-256 beginning `3bb4d890`); the Census Bureau's ASEC public-use bundles for 2023, 2024, and 2025 (the 2025 bundle, `asecpub25csv.zip`, has SHA-256 beginning `318845a2`); and the six OpenML datasets of Section 4.4, fetched by name at versions the loader asserts. The Federal Reserve replaced the 2022 extract at its published address on 9 October 2026 with a version re-based to 2025 dollars, so the file this paper used can no longer be downloaded from the agency. Each run directory's manifest records rows,

caps, seeds, and methods.

A Replication on the 2025 Survey of Consumer Finances

Every SCF experiment in the body of the paper uses the 2022 wave, and every design decision in the estimator and the harness was made with 2022 results in view. The Federal Reserve released the 2025 wave in October 2026 (Board of Governors of the Federal Reserve System, 2026a). We reran all three SCF experiments on it — the within-SCF task and both profiles of the population view — with the same code, locked environment, seeds, protocol, and receiver files (ASEC 2025 for the minimal profile, pooled ASEC 2023–2025 at production scale); only the donor wave changed. Before any method was fit on the new wave we committed a file fixing which of the paper's findings would count as replicated and by what rule. This appendix reports the result under those rules as written.

The 2025 file. The public summary extract holds 4,360 families in five implicates. Every predictor and target the two profiles use is present with the same codes; the one changed definition is race, where the Fed now folds the Middle Eastern or North African response into the white non-Hispanic class (Board of Governors of the Federal Reserve System, 2026a). (The separate Hispanic-origin questions dropped from the 2025 questionnaire never entered this variable.) The weights sum to 135.0 million families, against 135.0 million in the Fed's report, and the identity networth = fin + nfin − debt holds on every row. Pooled over implicates, all-family median and mean net worth are 216.1 and 1,232.0 thousand dollars, against published values of 215.9 and 1,241.5 thousand (gaps of +0.09 and −0.77 percent). The Fed's tables use internal data; the public file is disclosure-limited and omits the families whose net worth reaches the Forbes 400 threshold, seven in each wave (Board of Governors of the Federal Reserve System, 2026b). The 2022 public file's gaps are −0.10 and −0.40 percent.

Dollars. The 2025 extract is in 2025 dollars. On the same day the Fed replaced the 2022 extract at its published address with a copy re-based to 2025 dollars (every dollar column multiplied by the CPI-U-RS ratio 4789/4376; weights and codes unchanged). The paper's 2022 runs used the earlier, 2022-dollar file, and the repository's loader now accepts only files it can identify by hash, so the 2022 results regenerate regardless of which copy is downloaded. The comparisons below are scale-free or within-wave ratios. Dollar levels in the verdict tables are in each wave's own dollars.

Rules. Within the SCF, a claim that method B is worse than method A is met when B is worse on average and on at least eight of ten paired seeds. On the population view the noise unit is the sampling floor's seed-to-seed standard deviation: a difference is met beyond two floor standard deviations (with eight seeds agreeing), and a tie is met within one. Where the paper states an effect size, it holds if the 2025 effect is within a factor of two of the 2022 effect. Each rule is also applied to the 2022 runs; 2 of the paper's 2022 claims do not meet these rules on the 2022 data themselves, and for those the best possible 2025 verdict is inconclusive. The rules name the methods as the paper did when they were frozen. The candidate is the full estimator. The ungated forest is the `microimpute` forest of Table 1, an external baseline that has no gate but also chains its targets and draws through its own sampler. It is not the gates-off ablation of Section 3.2, which was added after the verdicts were scored.

Verdicts. Tables 9–11 list all 27 findings. Of the 8 that the abstract or conclusion states: 6 replicated, 2 inconclusive, 0 not replicated. Of the 19 others: 10 replicated, 2 replicated in direction with an effect smaller than the band allows, 2 inconclusive, 5 not replicated.

What replicated. The weighting result holds. Fit unweighted, the estimator is 4.29 times worse on net-worth W_1 within the SCF (6.10 in 2022), on 10 of ten seeds, and on the population view its imputed net-worth q99 is 1.79 and 1.78 times the holdout's (1.93 and 2.25 in 2022). Weighting is again the largest of the three ablation effects. Chaining again binds only at scale. With two targets the unchained ablation's AUC sits +0.36 floor standard deviations from the candidate's; on the four-component balance sheet unchaining costs 4.9 AUC points (5.8 in 2022), on 9 of ten seeds, and the ungated forest loses 5.9 points. The classifier separates the unconditional marginal draw as sharply as before (AUC 0.968 against the candidate's 0.757 at scale), and within the SCF the same draw again posts the best pinball loss on both targets. That last result is a property of the metric: the pinball loss reads the imputed quantiles unweighted and scores them against weighted truth. Read under receiver weights, the candidate's and the marginal draw's net-worth losses in 2025 are 531,304 and 531,697 (as coded, 691,571 and 531,502), and the marginal draw is the lower on 4 of ten seeds, not 10; the 2022 wave shows the same (4 of ten, not 10). The body's pinball columns are left as published. Hot deck stays within one floor standard deviation of the candidate on AUC in both profiles (+0.51 and -0.16). The candidate still understates the extreme tail: its net-worth q99 ratio is 0.642 and 0.630.

What did not. *Hot deck's tail at scale* (F4c, F5b). In 2022 hot deck matched the candidate on the net-worth tail at scale: a q99 ratio of 0.747 against the candidate's 0.767, and a W_1/sd of 0.024 against 0.027. In 2025 its q99 ratio at scale is 2.85 and its W_1/sd 0.103, against the candidate's 0.630 and 0.048. Donor reuse produces the q99 inflation. On 8 of the ten seeds, one donor above the holdout's q99 is matched to between 1.4 and 2.7 percent of receiver weight. A donor whose receivers carry more than one percent of the weight, by more than any single receiver's share, puts the imputed q99 at or above its own net worth, so that match alone lifts the imputed q99 above the holdout's. On the remaining seeds the largest such share is 0.04 percent, and in 2022 it never exceeded 0.09 percent. In the minimal profile hot deck's mean q99 ratio is 1.10, which looks like a correct tail and is not one: 1 of the ten seeds is inflated the same way, by a donor carrying 2.1 percent of receiver weight. The body's statement that hot deck slightly beats the candidate on the tail at scale holds for 2022 only.

The candidate as closest to the floor (F3c). At scale hot deck's mean AUC is 0.754, below the candidate's 0.757, so the candidate is not the lowest among itself and the baselines. The gap is -0.16 floor standard deviations: on this wave the candidate and hot deck are tied for closest to the floor.

Unchaining and the tail (F2c). At scale the unchained ablation's net-worth W_1/sd is 0.062 against the candidate's 0.048, +2.74 floor standard deviations apart; 3 of the 4 ties behind "essentially unchanged" were met.

Gating and debt within the SCF (G1). The 2022 runs showed no gating advantage on the debt zero share. In 2025 the ungated forest's error is 0.027 against the candidate's 0.017, and it is the worse of the two on 8 of ten seeds. The 2022 tie did not recur; the difference that replaced it favours the gated estimator.

Smaller effects. Unweighted is 1.49 times worse on debt W_1 within the SCF, against 2.48 in 2022. On the population view its tail W_1/sd is 3.12 and 1.69 times the candidate's, against 5.08 and 4.94; the body's "five times" does not hold at scale. Part of that is the candidate's own tail: its W_1/sd at scale rose from 0.027 to 0.048.

Inconclusive. Each inconclusive primary finding is a claim that several ties hold at once, and in each, one tie fell between one and two floor standard deviations. For the unweighted ablation (F1a), 5 of 6 ties were met: at scale its AUC is 0.735 against the candidate's 0.757 (-1.12 floor standard deviations; the unweighted fit is the harder to tell from the holdout). For hot deck (F4a), 3 of 4 were met: at scale its energy distance is 0.089 against 0.085

(+1.19). The claim that the ungated forest is worse than the candidate on net-worth W_1 within the SCF stayed within seed noise, as it was in 2022 (6 of ten seeds in 2025, 6 in 2022). The claim that removing chaining raises net-worth W_1 within the SCF met its rule in 2025 (a factor of 1.23, on 8 of ten seeds) but not in 2022 (7 of ten), so under the cap stated above it stays inconclusive.

Outside the rules. Energy distance fell for every method (the candidate's from 0.139 to 0.098 in the minimal profile and from 0.118 to 0.085 at scale). The criteria anticipated a shift shared by all methods, because SCF 2025 incomes refer to the same year as the ASEC 2025 receiver's; we did not test that explanation. The paper makes no claim about the net-worth q90 ratio. On this wave the candidate understates it at scale: 0.751, against 1.000 in 2022 and a floor of 0.991.

Method rankings. Ranking the eight displayed methods in each column of the three SCF tables, 10 of 15 rankings replicated under the pre-registered rank-correlation rule, 4 partly, and 1 not (debt zero-share error within the SCF, where the candidate, its ablations, the ungated forest, hot deck, and the marginal draw sit within a seed standard deviation of one another in both waves). The best method is unchanged in 10 columns.

Diagnostics. Rerunning the 2022 experiments in the environment used for 2025 reproduced the committed results exactly: 130 of 130 displayed table cells match on all ten seeds. Code, environment, and seeds are therefore held fixed between the two columns of the tables below; the donor wave, and with it the dollar basis, is what differs. Rerunning the 2022 wave in 2025 dollars left every 2022 status unchanged except F2c. (On the population view that rerun also shifts the donor's income predictors against the receiver's, whose incomes are not re-based.) The same arm shows that the forest-based methods are not exactly invariant to the unit of account: a pure change of units within the SCF leaves OLS and the marginal draw unchanged and moves the forest-based methods' ten-seed W_1 means by up to 7.3 percent. One new wave is one more draw. Where 2022 and 2025 disagree, these runs cannot say which is the unusual year.

Arms added later. The gates-off ablations, the net-worth-derived variants, and the weighted random and constrained hot decks joined the method surface after this replication was scored. Their rows in Tables 12–14 come from runs on the same splits and inputs. They are outside the pre-registered rules, and no verdict above uses them; where

the text above speaks of the estimator's ablations or of hot deck, it means the weights-off and chaining-off ablations and the nearest-neighbour hot deck. The added rows repeat the patterns of the body's tables. The gates-off ablation is the control that G1 lacked: G1 compared the full estimator with the `microimpute` forest, which differs from it in more than the gate. With the estimator's own gates off, its debt zero-share error is several times its error with them on, in both waves (Tables 2 and 12). The weighted random hot deck shows none of the tail inflation that donor reuse produces in the nearest-neighbour hot deck at scale, and understates the tail as it does on the earlier wave.

ID	Finding	2022	2025	Verdict
W1*	Fit unweighted, the same estimator is six times worse on net-worth W1.	829,158 vs 135,871; ratio 6.10; worse on 10/10 seeds	711,569 vs 165,931; ratio 4.29; worse on 10/10 seeds	Replicated
W2	Unweighted is 2.5 times worse on debt W1.	32,247 vs 13,001; ratio 2.48; worse on 10/10 seeds	18,537 vs 12,412; ratio 1.49; worse on 9/10 seeds	Replicated in direction, magnitude differs (smaller)
W3a	Weighting is the largest of the three ablation effects on net-worth W1.	deltas: unweighted +693,287; ungated forest +98,496; unchained +22,735	deltas: unweighted +545,638; ungated forest +29,485; unchained +37,883	Replicated
W3b	The ungated, unchained forest is worse than the candidate on net-worth W1 (1.7 times).	234,367 vs 135,871; ratio 1.72; worse on 6/10 seeds	195,416 vs 165,931; ratio 1.18; worse on 6/10 seeds	Inconclusive
W3c	Removing chaining raises net-worth W1 by 17 percent.	158,606 vs 135,871; ratio 1.17; worse on 7/10 seeds	203,814 vs 165,931; ratio 1.23; worse on 8/10 seeds	Inconclusive
W4a	Hot deck nearly ties the candidate on debt W1.	13,770 vs 13,001; ratio 1.06; hot deck worse on 5/10 seeds, better on 5/10	12,741 vs 12,412; ratio 1.03; hot deck worse on 5/10 seeds, better on 5/10	Replicated
W4b	Hot deck is 3.7 times worse on net-worth W1.	509,013 vs 135,871; ratio 3.75; worse on 9/10 seeds	434,090 vs 165,931; ratio 2.62; worse on 9/10 seeds	Replicated
W5	OLS is catastrophic on both targets.	OLS W1 is 12.0x (debt) and 26.2x (net worth) the candidate's; highest on both	OLS W1 is 10.1x (debt) and 16.3x (net worth) the candidate's; highest on both	Replicated
W6	Linear quantile regression leaves 15 percent of the debt zero mass unaccounted for, against at most 2 percent for forest-based methods.	quantile regression 0.151; forest-based 0.012 to 0.020	quantile regression 0.157; forest-based 0.017 to 0.027	Replicated
W7	The unconditional weighted marginal draw has the best pinball loss on both targets.	debt: marginal 48,277, candidate 49,201, lowest: marginal draw; networth: marginal 443,554, candidate 551,317, lowest: marginal draw	debt: marginal 52,793, candidate 53,883, lowest: marginal draw; networth: marginal 531,502, candidate 691,571, lowest: marginal draw	Replicated
G1	Gating shows no advantage on SCF debt zero share.	ungated forest 0.018 vs candidate 0.020; forest worse on 4/10 seeds, better on 6/10	ungated forest 0.027 vs candidate 0.017; forest worse on 8/10 seeds, better on 2/10	Not replicated

Table 9: Replication verdicts, within-SCF wealth task. Each cell gives the numbers the rule was applied to. Dollar amounts are in each wave's own dollars; ratios and seed counts are comparable across waves. An asterisk marks a finding the abstract or conclusion states.

ID	Finding	2022	2025	Verdict
F0	The sampling floor behaves as designed.	minimal: AUC 0.496, energy 0.009, coverage 0.987, q99 ratio 1.10; scale: AUC 0.499, energy 0.012, coverage 0.986, q99 ratio 1.10	minimal: AUC 0.506, energy 0.010, coverage 0.991, q99 ratio 0.99; scale: AUC 0.510, energy 0.013, coverage 0.990, q99 ratio 0.99	Replicated
F1a*	The unweighted ablation ties the candidate on energy distance, coverage, and C2ST AUC in both profiles.	largest gap 0.72 floor sd (coverage, scale: 0.971 vs 0.969); 6/6 ties met	largest gap 1.12 floor sd (AUC, scale: 0.735 vs 0.757); 5/6 ties met	Inconclusive
F1b*	Its imputed net-worth q99 is about twice the holdout's (1.93 and 2.25).	minimal: 1.93 vs candidate 0.70 (6.1 floor sd; 10/10 seeds); scale: 2.25 vs candidate 0.77 (7.3 floor sd; 10/10 seeds)	minimal: 1.79 vs candidate 0.64 (7.5 floor sd; 10/10 seeds); scale: 1.78 vs candidate 0.63 (7.5 floor sd; 10/10 seeds)	Replicated
F1c	Its tail W1/sd is five times the candidate's.	minimal: 0.159 vs 0.031, ratio 5.08 (10/10 seeds); scale: 0.133 vs 0.027, ratio 4.94 (10/10 seeds)	minimal: 0.125 vs 0.040, ratio 3.12 (9/10 seeds); scale: 0.081 vs 0.048, ratio 1.69 (8/10 seeds)	Replicated in direction, magnitude differs (smaller)
F2a*	With two targets the unchained ablation is indistinguishable from the candidate.	0.742 vs 0.732; gap 0.40 floor sd	0.727 vs 0.719; gap 0.36 floor sd	Replicated
F2b*	On the four-component balance sheet, fitting targets independently costs 5.8 AUC points.	0.819 vs 0.761; gap +0.058 (2.70 floor sd); higher on 9/10 seeds	0.806 vs 0.757; gap +0.049 (2.54 floor sd); higher on 9/10 seeds	Replicated
F2c	Unchaining leaves the marginal and tail statistics essentially unchanged at scale.	largest gap 0.76 floor sd (coverage, scale: 0.967 vs 0.969); 4/4 ties met	largest gap 2.74 floor sd (W1/sd, scale: 0.062 vs 0.048); 3/4 ties met	Not replicated
F2d	The ungated forest loses 5.6 AUC points at scale.	0.817 vs 0.761; gap +0.056 (2.63 floor sd); higher on 10/10 seeds	0.816 vs 0.757; gap +0.059 (3.03 floor sd); higher on 9/10 seeds	Replicated

Table 10: Replication verdicts, population view: the floor, weighting, and chaining. "Floor sd" is the seed-to-seed standard deviation of the sampling-floor row. An asterisk marks a finding the abstract or conclusion states.

ID	Finding	2022	2025	Verdict
F3a*	At scale the classifier separates the unconditional marginal draw almost perfectly (AUC 0.97 against the candidate's 0.76).	0.968 vs 0.761; gap +0.207 (9.68 floor sd); higher on 10/10 seeds	0.968 vs 0.757; gap +0.211 (10.90 floor sd); higher on 10/10 seeds	Replicated
F3b	Conditioning richness makes the test decisive: the marginal draw's AUC rises from the minimal profile to scale.	0.848 (minimal) to 0.968 (scale): rise +0.120	0.845 (minimal) to 0.968 (scale): rise +0.123	Replicated
F3c	The candidate stays closest to the floor.	minimal: candidate 0.732, lowest baseline 0.753; scale: candidate 0.761, lowest baseline 0.768	minimal: candidate 0.719, lowest baseline 0.731; scale: candidate 0.757, lowest baseline 0.754	Not replicated
F4a*	Hot deck is within noise of the candidate on energy distance and coverage in both profiles.	largest gap 0.71 floor sd (coverage, scale: 0.971 vs 0.969); 4/4 ties met	largest gap 1.19 floor sd (energy, scale: 0.089 vs 0.085); 3/4 ties met	Inconclusive
F4b*	Hot deck is within 2.1 and 0.7 AUC points of the candidate.	largest gap 0.86 floor sd (AUC, minimal: 0.753 vs 0.732); 2/2 ties met	largest gap 0.51 floor sd (AUC, minimal: 0.731 vs 0.719); 2/2 ties met	Replicated
F4c	At scale hot deck slightly beats the candidate on tail W1/sd.	hot deck 0.024 vs candidate 0.027 (-0.0028, 0.90 floor sd)	hot deck 0.103 vs candidate 0.048 (+0.0550, 10.50 floor sd)	Not replicated
F5a	The candidate understates the extreme tail (net-worth q99 ratio 0.70 and 0.77).	minimal: 0.70, below 1 on 10/10 seeds; scale: 0.77, below 1 on 9/10 seeds	minimal: 0.64, below 1 on 10/10 seeds; scale: 0.63, below 1 on 10/10 seeds	Replicated
F5b	Hot deck does not fix the tail either, while the marginal draw sits near the floor (read from the 2022 tables, not the text).	minimal: hot deck 0.76 vs candidate 0.70 (0.33 floor sd); marginal 1.09 vs floor 1.10 (0.05 floor sd); scale: hot deck 0.75 vs candidate 0.77 (0.10 floor sd); marginal 1.12 vs floor 1.10 (0.07 floor sd)	minimal: hot deck 1.10 vs candidate 0.64 (3.01 floor sd); marginal 0.99 vs floor 0.99 (0.02 floor sd); scale: hot deck 2.85 vs candidate 0.63 (14.56 floor sd); marginal 1.00 vs floor 0.99 (0.09 floor sd)	Not replicated

Table 11: Replication verdicts, population view, continued: the marginal draw, hot deck, and the tail.

Method	Debt pinball	Debt W_1	Debt zero-share err.	Net worth pinball	Net worth W_1
microcosm-fit	53,883 (2,516)	**12,412** (4,467)	**0.017** (0.022)	691,571 (65,700)	**165,931** (90,856)
– unweighted	54,061 (2,224)	18,537 (7,279)	0.021 (0.018)	853,940 (69,647)	711,569 (313,876)
– unchained	53,883 (2,516)	**12,412** (4,467)	**0.017** (0.022)	693,429 (46,890)	203,814 (57,212)
– ungated	53,523 (2,060)	14,867 (6,444)	0.095 (0.033)	708,402 (61,593)	189,257 (84,192)
– ungated, unchained	53,523 (2,060)	14,867 (6,444)	0.095 (0.033)	692,310 (54,527)	202,673 (48,477)
QRF (microimpute)	55,184 (1,927)	13,575 (3,675)	0.027 (0.011)	689,111 (21,625)	195,416 (62,104)
OLS	69,765 (6,754)	125,966 (31,799)	0.230 (0.013)	1,116,887 (197,719)	2,703,593 (964,714)
Quantile regression	54,264 (1,963)	21,064 (5,644)	0.157 (0.015)	672,364 (41,739)	235,061 (46,789)
NND hot deck (py-statmatch)	53,598 (2,172)	12,741 (3,353)	0.020 (0.020)	786,929 (52,769)	434,090 (251,013)
Weighted random hot deck	53,434 (2,215)	13,775 (4,073)	0.019 (0.013)	635,846 (43,349)	198,962 (53,007)
Constrained hot deck	53,535 (2,220)	13,418 (4,675)	0.019 (0.018)	793,722 (53,520)	389,354 (164,775)
Weighted marginal draw	**52,793** (2,339)	13,178 (3,605)	0.019 (0.010)	**531,502** (32,912)	215,488 (90,663)

Table 12: Table 2 on the 2025 wave (2025 dollars).

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.010 (0.004)	0.991 (0.005)	0.506 (0.022)	0.018 (0.005)	0.990 (0.153)
microcosm-fit	**0.098** (0.019)	0.964 (0.009)	0.719 (0.009)	0.040 (0.010)	0.642 (0.122)
– unweighted	0.099 (0.019)	**0.966** (0.009)	**0.712** (0.010)	0.125 (0.068)	1.79 (0.726)
– unchained	0.098 (0.019)	0.964 (0.010)	0.727 (0.012)	0.046 (0.017)	0.728 (0.170)
– ungated	0.098 (0.019)	0.965 (0.012)	0.722 (0.013)	0.040 (0.009)	0.637 (0.112)
– ungated, unchained	0.098 (0.019)	0.965 (0.012)	0.732 (0.013)	0.044 (0.014)	0.728 (0.324)
QRF (microimpute)	0.100 (0.018)	0.961 (0.013)	0.764 (0.013)	0.039 (0.008)	0.699 (0.176)
OLS	0.116 (0.025)	0.759 (0.067)	0.983 (0.004)	0.351 (0.130)	0.676 (0.215)
Quantile regression	0.101 (0.020)	0.957 (0.012)	0.907 (0.007)	0.060 (0.010)	0.462 (0.091)
NND hot deck (py-statmatch)	0.102 (0.020)	0.965 (0.009)	0.731 (0.011)	0.058 (0.036)	1.10 (0.808)
Weighted random hot deck	0.103 (0.020)	0.960 (0.011)	0.744 (0.013)	0.072 (0.016)	0.457 (0.063)
Weighted marginal draw	0.101 (0.019)	0.940 (0.017)	0.845 (0.010)	**0.021** (0.008)	**0.994** (0.181)

Table 13: Table 7 on the 2025 wave: the population view, minimal profile, with the SCF 2025 donor.

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.013 (0.003)	0.990 (0.006)	0.510 (0.019)	0.018 (0.005)	0.990 (0.153)
microcosm-fit	0.085 (0.015)	0.976 (0.005)	0.757 (0.048)	0.048 (0.016)	0.630 (0.165)
– unweighted	0.086 (0.015)	0.976 (0.004)	0.735 (0.013)	0.081 (0.042)	1.78 (0.745)
– unchained	0.085 (0.015)	0.973 (0.006)	0.806 (0.014)	0.062 (0.059)	0.692 (0.195)
– ungated	**0.085** (0.015)	0.973 (0.006)	0.750 (0.012)	0.045 (0.011)	0.602 (0.111)
– ungated, unchained	0.085 (0.015)	0.971 (0.006)	0.817 (0.013)	0.048 (0.013)	0.592 (0.079)
– net worth derived	0.085 (0.015)	**0.976** (0.004)	**0.732** (0.008)	0.043 (0.013)	0.679 (0.204)
– unchained, net worth derived	0.085 (0.015)	0.976 (0.005)	0.743 (0.008)	0.043 (0.013)	0.674 (0.195)
QRF (microimpute)	0.092 (0.017)	0.968 (0.009)	0.816 (0.011)	0.030 (0.007)	0.916 (0.261)
OLS	0.103 (0.018)	0.902 (0.025)	0.993 (0.002)	0.349 (0.130)	0.663 (0.203)
Quantile regression	0.087 (0.015)	0.960 (0.011)	0.965 (0.004)	0.064 (0.011)	0.425 (0.074)
NND hot deck (py-statmatch)	0.089 (0.016)	0.975 (0.005)	0.754 (0.009)	0.103 (0.043)	2.85 (1.30)
Weighted random hot deck	0.087 (0.015)	0.971 (0.006)	0.769 (0.011)	0.055 (0.014)	0.517 (0.067)
Weighted marginal draw	0.089 (0.015)	0.930 (0.009)	0.968 (0.003)	**0.019** (0.007)	**1.00** (0.147)

Table 14: Table 8 on the 2025 wave: the population view, production-scale profile, with the SCF 2025 donor.

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